

Program-4: Integrate Git with Docker for Source-Controlled Applications Builds

Project Folder Structure

program-4/

- Dockerfile
- index.js
- package.json
- package-lock.json
- .gitignore

1. Create package.json (Node project)

npm init -y

2. Create index.js (Express app)

nano index.js

```
GNU nano 7.2 index.js
const express = require("express");
const app = express();
const PORT = 3000;

app.get("/", (req, res) => {
  res.send("Hello from Git + Docker Integration!");
});

app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

3. Install Express dependency

npm install express

4. Create Dockerfile

nano Dockerfile

```
GNU nano 7.2 Dockerfile
# Stage 1 - Build Stage
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

# Stage 2 - Runtime Stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node", "index.js"]
```

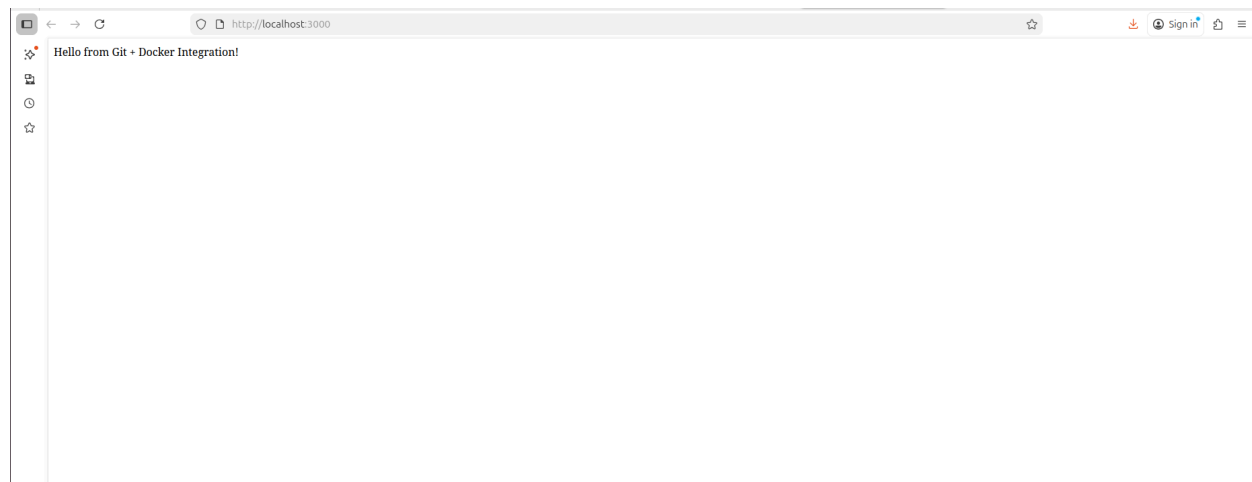
5. Build and run Docker image locally

5.1 Build image

`docker build -t program4-app .`

5.2 Run container

`docker run -p 3000:3000 program4-app`



6. Create a new GitHub repository (online)

- Login to GitHub in browser.
- At the top-right, click + icon → New repository.
- Repository name: e.g. program-4
- Visibility: Public / Private.

- Do NOT add README, .gitignore, or license.
- Click Create repository.

7. Create a Personal Access Token (PAT)

- On GitHub, click your profile picture → Settings.
- Scroll left side to Developer settings.
- Click Personal access tokens → Tokens (classic).
- Click Generate new token → Generate new token (classic).
- Give a name (e.g. devops-lab-token), set expiry (e.g. 30 days), select scopes.
- Click Generate token.
- Copy the token and keep it somewhere safe.

8. Initialize Git in your local program-4 folder

git init

git status

Now add all files and commit:

git add .

git commit -m "Initial commit"

Ensure branch is main:

git branch -M main

```

1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1rv24mc120_yashwanth_c/devOps/program4/.git/
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile
    index.js
    node_modules/
    package-lock.json
    package.json

nothing added to commit but untracked files present (use "git add" to track)
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git add .
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git commit -m "Initial commit"
[master (root-commit) 2d9d108] Initial commit
589 files changed, 61783 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js

```

9. Connect local repo to GitHub remote

git remote add origin https://github.com/<your-username>/<repo-name>.git

10. Push code to GitHub

git push -u origin main

```

1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git remote add origin https://github.com/Yashwanth142003/program-4.git
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git remote -v
origin https://github.com/Yashwanth142003/program-4.git (fetch)
origin https://github.com/Yashwanth142003/program-4.git (push)
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/Yashwanth142003/program-4.git'
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git branch
* master
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git add .
git commit -m "Initial commit"
On branch master
nothing to commit, working tree clean
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git branch -M main
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program4$ git push -u origin main
Username for 'https://github.com': Yashwanth142003
Password for 'https://Yashwanth142003@github.com':
Enumerating objects: 678, done.
Counting objects: 100% (678/678), done.
Delta compression using up to 12 threads
Compressing objects: 100% (627/627), done.
Writing objects: 100% (678/678), 669.86 KiB | 10.98 MiB/s, done.
Total 678 (delta 121), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (121/121), done.
To https://github.com/Yashwanth142003/program-4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

```

11. Verify on GitHub

Go back to the GitHub repo page in the browser and refresh.

